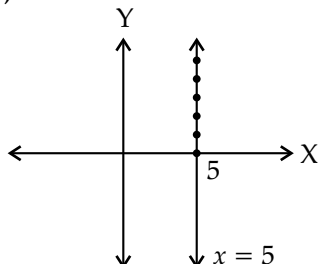


$$\begin{aligned}\therefore u + v + w &= -\frac{3}{2} + 0 - \frac{1}{2} \\ &= -\frac{4}{2} \\ &= -2\end{aligned}$$

53. Option (d) is correct.



$x = 5$ represents a line that is parallel to y -axis and cuts the x -axis at 5 units from the origin.

It is given that $x = 5$, this implies that the y -coordinate is zero.

Here the infinite number of points i.e., $(5, y)$ represent the equation $x = 5$ on the xy -plane.

54. Option (b) is correct.

We know the plane equation is $ax + by + cz = d$.

Given the plane equation is

$$2x - 3y + 6z + 4 = 0$$

$$\Rightarrow 2x - 3y + 6z = -4$$

Comparing with plane equation we get

$$a = 2, b = -3, c = 6, d = -4$$

$$\begin{aligned}\therefore \sqrt{a^2 + b^2 + c^2} &= \sqrt{(2)^2 + (-3)^2 + (6)^2} \\ &= \sqrt{4 + 9 + 36} = \sqrt{49} \\ &= 7\end{aligned}$$

Here we need to find the direction cosine of a normal to the plane i.e., $\langle l, m, n \rangle$

Direction cosines of normal to plane

$$\begin{aligned}l &= \frac{a}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{2}{\sqrt{49}} = \frac{2}{7}\end{aligned}$$

$$\begin{aligned}m &= \frac{b}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{-3}{\sqrt{49}} = \frac{-3}{7}\end{aligned}$$

$$\begin{aligned}n &= \frac{c}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{6}{\sqrt{49}} = \frac{6}{7}\end{aligned}$$

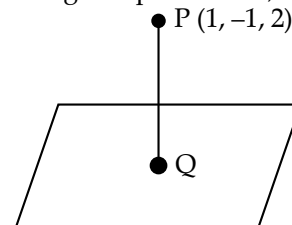
$$\therefore \langle l, m, n \rangle = \left\langle \frac{2}{7}, \frac{-3}{7}, \frac{6}{7} \right\rangle$$

$$\begin{aligned}\therefore 49(7l^2 + m^2 - n^2) &= \\ &= 49 \left[7 \left(\frac{2}{7} \right)^2 + \left(\frac{-3}{7} \right)^2 - \left(\frac{6}{7} \right)^2 \right] \\ &= 49 \left(\frac{28}{49} + \frac{9}{49} - \frac{36}{49} \right) \\ &= 49 \left(\frac{37 - 36}{49} \right) \\ &= 1\end{aligned}$$

55. Option (c) is correct.

Let the line from point $P(1, -1, 2)$ and meet the plane at point Q .

Direction ratios of the line from the point $(1, -1, 2)$ to the given plane is $\langle 3, 2, 2 \rangle$



So the equation of the line passing through P and with direction ratios will be:

$$\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-2}{2} = \lambda$$

Coordinates of any points on the line PQ are

$$x = 3\lambda + 1, y = 2\lambda - 1, z = 2\lambda + 2$$

Now, since Q lies on the plane so it must satisfy the equation of the plane.

$$\text{i.e., } x + 2y + 3z = 18$$

$$\therefore 3\lambda + 1 + 4\lambda - 2 + 6\lambda + 6 = 18$$

$$\Rightarrow 13\lambda + 5 = 18$$

$$\Rightarrow \lambda = 1$$

So, the coordinates of Q are:

$$(3 + 1, 2 - 1, 2 + 2) = (4, 1, 4)$$

56. Option (d) is correct.

Given that plane passing through $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$.

First we will find the equation of plane

$$\vec{a} = \hat{i}, \vec{b} = \hat{j}, \vec{c} = \hat{k}$$

$$\text{Let } \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0$$

$$\therefore \vec{b} - \vec{a} = \hat{j} - \hat{i} \text{ and } \vec{c} - \vec{a} = \hat{k} - \hat{i}$$